



Hyperthyroidism

(Thyrotoxicosis, including Graves Disease)

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Hyperthyroidism is characterized by hypermetabolism and elevated serum levels of free thyroid hormones. Symptoms include palpitations, fatigue, weight loss, heat intolerance, anxiety, and tremor. Diagnosis is based on serum thyroid function tests. Treatment depends on cause.

Hyperthyroidism affects about 1% of people in the United States. It can occur at any age but is more common in women between the ages of 20 and 50 years ([1](#)).

Hyperthyroidism can be classified on the basis of thyroid radioactive iodine uptake and the level of circulating thyroid hormones (see table [Results of Thyroid Function Tests in Various Clinical Situations](#)).

General reference

1. [Lee SY, Pearce EN](#). Hyperthyroidism: A Review. *JAMA*. 2023;330(15):1472-1483. doi:10.1001/jama.2023.19052

Etiology of Hyperthyroidism

Hyperthyroidism may result from increased [synthesis and secretion of thyroid hormones](#) (thyroxine [T4] and triiodothyronine [T3]) from the thyroid, caused by thyroid-stimulating immunoglobulin (TSI) or by autonomous thyroid hyperfunction. It can also result from excessive release of thyroid hormone from the thyroid without increased synthesis. Such release is commonly caused by the destructive changes of various types of thyroiditis.

The **most common causes** of hyperthyroidism include:

- Graves disease
- Multinodular goiter
- Single, autonomous, hyperfunctioning "hot" nodule
- Thyroiditis

Graves disease (toxic diffuse goiter), the most common cause of hyperthyroidism, is characterized by hyperthyroidism and one or more of the following:

- Goiter
- [Exophthalmos](#)
- Infiltrative dermopathy

Graves disease is caused by an autoantibody (TSI) against the thyroid receptor for thyroid-stimulating hormone (TSH); unlike most autoantibodies, which are inhibitory, this autoantibody is stimulatory, thus causing continuous synthesis and secretion of excess T4 and T3. Graves disease (like [Hashimoto thyroiditis](#)) sometimes occurs with other autoimmune disorders, including type 1 [diabetes mellitus](#), [vitiligo](#), premature graying of hair, pernicious anemia, systemic rheumatic disease, and [polyglandular deficiency syndrome](#). Heredity increases the risk of Graves disease, although the genes involved are unknown.

The pathogenesis of infiltrative ophthalmopathy (responsible for the exophthalmos in Graves disease) is poorly understood but may result from immunoglobulins directed to the TSH receptors in the orbital fibroblasts and fat that result in release of proinflammatory cytokines, inflammation, and accumulation of glycosaminoglycans. Ophthalmopathy may also occur before the onset of hyperthyroidism or as late as 20 years afterward and frequently worsens or abates independently of the clinical course of hyperthyroidism. Typical ophthalmopathy in the presence of normal thyroid function is called euthyroid Graves disease (see photo [Eye Manifestations of Graves Disease—Exophthalmos](#)).

Toxic solitary or multinodular goiter (Plummer disease) sometimes results from TSH receptor gene mutations causing continuous thyroid activation. Patients with toxic nodular goiter have none of the autoimmune manifestations or circulating antibodies observed in patients with Graves disease. Also, in contrast to Graves disease, toxic solitary and multinodular goiters usually do not remit.

Inflammatory thyroid disease (thyroiditis) includes subacute granulomatous thyroiditis, [Hashimoto thyroiditis](#), and [silent lymphocytic thyroiditis](#), a variant of Hashimoto thyroiditis. Hyperthyroidism is more common in [subacute granulomatous thyroiditis](#) and results from destructive changes in the gland and release of stored hormone, not from increased synthesis. Hypothyroidism may follow.

Inappropriate TSH secretion is a rare cause. Patients with primary hyperthyroidism have undetectable TSH levels. However, TSH levels are detectable in patients with a TSH-secreting anterior pituitary adenoma or with pituitary resistance to thyroid hormone, in whom TSH levels are high. The TSH produced in both of these pituitary disorders is biologically more active than normal TSH. An increase in the alpha-subunit of TSH in the blood (helpful in differential diagnosis) occurs in patients with a TSH-secreting pituitary adenoma.

Medication-induced hyperthyroidism can result from [amiodarone](#), checkpoint inhibitors used in cancer therapy, [alemtuzumab](#) used in treatment of multiple sclerosis, or interferon-alfa, which may induce thyroiditis with hyperthyroidism and other thyroid disorders. Although more commonly causing hypothyroidism, [lithium](#) can rarely cause hyperthyroidism. Patients receiving these medications should be closely monitored.

Thyrotoxicosis factitia is hyperthyroidism resulting from intentional or accidental overingestion of thyroid hormone.

Excess iodine ingestion causes hyperthyroidism with a low thyroid radioactive iodine uptake. It most often occurs in patients with underlying nontoxic nodular goiter (especially older patients) who are given medications that contain iodine (eg, [amiodarone](#)) or who undergo radiologic studies using iodine-rich contrast agents. The etiology may be that the excess iodine provides substrate for functionally autonomous (ie, not under TSH regulation) areas of the thyroid to produce hormone. Hyperthyroidism usually persists as long as excess iodine remains in the circulation.

Hydatidiform mole (molar pregnancy) and choriocarcinoma produce high levels of serum human chorionic gonadotropin (hCG), a weak thyroid stimulator. Levels of hCG are highest during the first trimester of pregnancy and result in the decrease in serum TSH and mild increase in serum free T4 sometimes observed at that time. The increased thyroid stimulation may be caused by increased levels of partially desialated hCG, an hCG variant that appears to be a more potent thyroid stimulator than more sialated hCG. Hyperthyroidism in molar pregnancy, choriocarcinoma, and [hyperemesis gravidarum](#) is transient; normal thyroid function resumes when the molar pregnancy is evacuated, the choriocarcinoma is appropriately treated, or the hyperemesis gravidarum abates.

Nonautoimmune autosomal dominant hyperthyroidism manifests during infancy. It results from mutations in the TSH receptor gene that produce continuous thyroid stimulation.

Metastatic thyroid cancer is a possible cause. Overproduction of thyroid hormone occurs rarely from functioning metastatic follicular carcinoma, especially in pulmonary metastases.

Struma ovarii develops when ovarian teratomas contain enough thyroid tissue to cause true hyperthyroidism. Radioactive iodine uptake occurs in the pelvis, and uptake by the thyroid is usually suppressed.

Risk factors associated with hyperthyroidism include: female sex, older age, and smoking ([1](#)).

Etiology reference

1. [Chaker L, Cooper DS, Walsh JP, Peeters RP](#). Hyperthyroidism. *Lancet*. 2024;403(10428):768-780. doi:10.1016/S0140-6736(23)02016-0

Pathophysiology of Hyperthyroidism

In hyperthyroidism, serum T3 usually increases more than does T4, probably because of increased secretion of T3 as well as conversion of T4 to T3 in peripheral tissues. In some patients, only T3 is elevated (T3 toxicosis).

T3 toxicosis may occur in any of the common disorders that cause hyperthyroidism, including Graves disease, multinodular goiter, and the autonomously functioning solitary thyroid nodule. If T3 toxicosis is untreated, the patient usually also develops laboratory abnormalities typical of hyperthyroidism (ie,

elevated T4 and increased iodine-123 uptake). The various forms of thyroiditis commonly have a hyperthyroid phase followed by a hypothyroid phase.

Symptoms and Signs of Hyperthyroidism

Most symptoms and signs are the same regardless of the cause. Exceptions include infiltrative ophthalmopathy and infiltrative dermopathy, which occur only in Graves disease.

Pearls & Pitfalls

- Older adults with hyperthyroidism may have symptoms similar to depression or dementia.

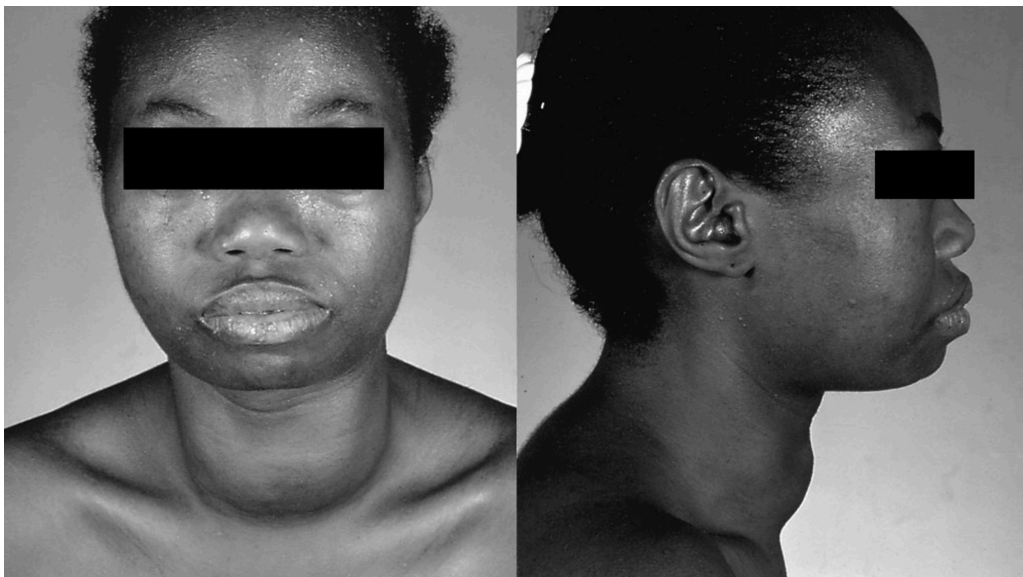
The clinical presentation may be dramatic or subtle. A goiter or nodule may be present.

Many common symptoms of hyperthyroidism are due to enhanced sensitivity to adrenergic hormones, such as nervousness, palpitations, hyperactivity, increased sweating, heat hypersensitivity, fatigue, increased appetite, weight loss, insomnia, weakness, and frequent bowel movements (occasionally diarrhea). Hypomenorrhea may be present.

Signs may include warm, moist skin, tremor, tachycardia, widened pulse pressure, and atrial fibrillation.

Graves Disease (Enlarged Thyroid Gland)

IMAGE



In this patient with Graves disease, the thyroid is visibly enlarged. Typically, the thyroid feels smooth, without discrete nodules, on palpation.

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Older patients, particularly those with toxic nodular goiter, may present atypically (apathetic or masked hyperthyroidism) with symptoms more akin to [depression](#) or [dementia](#). Most do not have exophthalmos or tremor. [Atrial fibrillation](#), [syncope](#), altered sensorium, [heart failure](#), and weakness are more likely. Symptoms and signs may involve only a single organ system.

Eye signs

Eye signs of hyperthyroidism include stare, eyelid lag, eyelid retraction, and mild conjunctival injection and are largely due to excessive adrenergic stimulation. They usually remit with successful treatment to normalize thyroid hormone levels.

[Infiltrative ophthalmopathy](#), also called thyroid eye disease (TED), is a more serious development and is specific to Graves disease. It can occur years before or after hyperthyroidism. It is characterized by orbital pain, lacrimation, irritation, photophobia, increased retro-orbital tissue, exophthalmos, and lymphocytic infiltration of the extraocular muscles, causing ocular muscle weakness that frequently leads to double vision.

Eye Manifestations of Graves Disease



Infiltrative Ophthalmopathy in Graves Disease

A patient with Graves disease manifests ocular signs, including widened palpebral fissures, lid retraction, strabismus, and exophthalmos.

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Eye Manifestations of Graves Disease—Exophthalmos

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Close-up View of Exophthalmos

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Eye Manifestations of Graves Disease—Inability to Close the Eyes

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Eye Manifestations of Graves Disease—Infraorbital Bags

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Infiltrative dermopathy

Infiltrative dermopathy, also called pretibial myxedema (a confusing term, because myxedema suggests hypothyroidism), is characterized by nonpitting infiltration by proteinaceous ground substance, usually

in the pretibial area. It rarely occurs in the absence of Graves ophthalmopathy. The lesion is often pruritic and erythematous in its early stages and subsequently becomes brawny. Infiltrative dermopathy may appear years before or after hyperthyroidism.

Manifestations of Pretibial Myxedema



Pretibial Myxedema

This photo shows a patient with pretibial myxedema characterized by non-pitting edema, thickened scaly skin, and hyperpigmentation.

Photo courtesy of Karen McKoy, MD.



Pretibial Myxedema (Mild)

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Pretibial Myxedema in a Patient With Graves Disease

By permission of the publisher. From Burman K, Becker K, Cytryn A, et al. In *Atlas of Clinical Endocrinology: Thyroid Diseases*. Edited by SG Korenman (series editor) and MI Surks. Philadelphia, Current Medicine, 1999.

Thyroid storm

Thyroid storm is an acute form of hyperthyroidism that results from untreated or inadequately treated severe hyperthyroidism. It is rare, occurring in patients with Graves disease or toxic multinodular goiter (a solitary toxic nodule is a less common cause and generally causes less severe manifestations). It may be precipitated by infection, trauma, surgery, embolism, diabetic ketoacidosis, or preeclampsia.

Thyroid storm causes abrupt florid symptoms of hyperthyroidism with one or more of the following: fever, marked weakness and muscle wasting, extreme restlessness with wide emotional swings, confusion, psychosis, coma, nausea, vomiting, diarrhea, and hepatomegaly with mild jaundice. The patient may present with tachycardia or cardiovascular collapse and shock. *Thyroid storm is a life-threatening emergency requiring prompt treatment.*

Diagnosis of Hyperthyroidism

- TSH measurement
- Free T4, plus either free T3 or total T3 measurement
- Sometimes radioactive iodine uptake

Diagnosis of hyperthyroidism is based on history, physical examination, and thyroid function tests.

Physical examination may reveal systolic hypertension, widened pulse pressure, resting tachycardia, and an irregularly irregular cardiac rhythm suggesting atrial fibrillation. Examine the thyroid for size, symmetry, tenderness (suggesting thyroiditis), nodules, and a systolic bruit (suggestive of Graves disease). Extremities, hair, and nails may reveal warm moist skin, warm extremities, pretibial myxedema (in Graves disease), fine hair, and onycholysis. Ophthalmologic findings may include lid retraction, lid lag, stare, conjunctival injection, and [proptosis](#). On neurologic examination, patients may have a fine high-frequency tremor of the hands, hyperreflexia with a brisk relaxation phase, or proximal muscle weakness.

Serum TSH measurement is the best test because TSH is suppressed in patients with hyperthyroidism except in the rare instance when the etiology is a TSH-secreting pituitary adenoma or pituitary resistance to the normal inhibition by thyroid hormone.

Free T4 is increased in hyperthyroidism. However, T4 can be falsely normal in true hyperthyroidism in patients with a severe systemic illness (similar to the falsely low levels that occur in [euthyroid sick syndrome](#)) and in T3 toxicosis. If the free T4 level is normal and TSH is low in a patient with subtle symptoms and signs of hyperthyroidism, then serum T3 should be measured to detect T3 toxicosis; an elevated level confirms that diagnosis.

The cause can often be diagnosed clinically (eg, the presence of signs specific to Graves disease). If not, radioactive iodine uptake by the thyroid may be measured by using iodine-123. When hyperthyroidism is due to hormone overproduction, radioactive iodine uptake by the thyroid is usually elevated (diffusely in Graves disease, focally in uninodular or multinodular goiter). When hyperthyroidism is due to thyroiditis, iodine ingestion, or overtreatment with thyroid hormones, radioactive iodine uptake is low.

TSH receptor antibodies can be measured to evaluate for Graves disease. Measurement is done in pregnant patients with a history of Graves disease during the third trimester of pregnancy to assess the risk of [neonatal Graves disease](#); TSH receptor antibodies readily cross the placenta to stimulate the fetal thyroid. Most patients with Graves disease have circulating antithyroid peroxidase antibodies, and fewer have antithyroglobulin antibodies.

Inappropriate TSH secretion is uncommon. The diagnosis is confirmed when hyperthyroidism occurs with elevated circulating free T4 and T3 concentrations and normal or elevated serum TSH.

If thyrotoxicosis factitia is suspected, serum thyroglobulin can be measured; it is usually low or low-normal—unlike in all other causes of hyperthyroidism.

Subclinical hyperthyroidism

Subclinical hyperthyroidism is low serum TSH in patients with normal serum free T4 and T3 and absent or minimal symptoms of hyperthyroidism.

Subclinical hyperthyroidism is far less common than [subclinical hypothyroidism](#).

Many patients with subclinical hyperthyroidism are taking [levothyroxine](#). The other causes of subclinical hyperthyroidism are the same as those for clinically apparent [hyperthyroidism](#).

Patients with serum TSH < 0.1 microU/mL (0.1 mU/L) have an increased incidence of [atrial fibrillation](#) (particularly older patients), reduced bone mineral density, increased fractures, and increased mortality. Patients with serum TSH that is only slightly below normal are less likely to have these complications.

Treatment of Hyperthyroidism

Treatment of hyperthyroidism depends on the cause ([1](#)) but may include the following:

- Radioactive iodine
- [Methimazole](#) or [propylthiouracil](#)
- Beta-blockers
- Iodine
- Thyroidectomy

Radioactive sodium iodine (iodine-131, radioiodine)

In the United States, iodine-131 is the most common treatment for hyperthyroidism. Radioiodine is often recommended as the treatment of choice for Graves disease and toxic nodular goiter in all

patients, including children. Epidemiologic studies report a very small increased risk of secondary malignancies with I-131 treatment, depending on the type of cancer and length of follow-up (2), so some clinicians use antithyroid medications instead. Dosage of iodine-131 is difficult to adjust because the response of the gland cannot be predicted; some clinicians give a standard dose of 8 to 15 millicurie. Others adjust the dose based on estimated thyroid size and the 24-hour uptake to provide a dose of 80 to 120 microcurie/g thyroid tissue.

When sufficient iodine-131 is given to cause euthyroidism, approximately 25 to 50% of patients become hypothyroid 1 year later, and the incidence continues to increase yearly. Thus, most patients eventually become hypothyroid and require thyroid hormone replacement. However, if smaller doses are used, incidence of recurrence is higher. Larger doses, such as 10 to 20 millicurie, often cause hypothyroidism within 6 months.

Radioactive iodine is not used during lactation because it can enter breast milk and cause hypothyroidism in the infant. It is not used during pregnancy because it crosses the placenta and can cause severe fetal hypothyroidism. There is no proof that radioiodine increases the incidence of tumors, leukemia, thyroid cancer, or birth defects in children born to previously hyperthyroid women who become pregnant after treatment was completed.

Methimazole and propylthiouracil

These antithyroid medications block thyroid peroxidase, decreasing the organification of iodide, and impair the coupling reaction. [Propylthiouracil](#) in high doses also inhibits the peripheral conversion of T4 to T3.

[Methimazole](#) is the preferred medication. The usual starting dose of methimazole is 5 to 20 mg orally 1 to 3 times a day. Normalization of TSH lags normalization of T4 and T3 levels by weeks. Therefore, when T4 and T3 levels normalize, the dosage is decreased to the lowest effective amount, usually methimazole 2.5 to 10 mg once a day in order to avoid inducing hypothyroidism. Control generally is achieved in 2 to 3 months. Maintenance doses of methimazole should be continued for ≥ 1 year depending on the clinical circumstances. Carbimazole, which is used widely in Europe but is unavailable in the United States, is rapidly converted to methimazole. The usual starting dose is similar to that of methimazole; maintenance dosage is 2.5 to 10 mg orally once a day or 2.5 to 5 mg twice a day.

Because of severe liver failure in some patients < 40 years old, especially children, [propylthiouracil](#) is recommended only in special situations (eg, during the first trimester of pregnancy, in thyroid storm). The usual starting dose of propylthiouracil is 100 to 150 mg orally every 8 hours. Rapid control can be achieved by increasing the dosage of propylthiouracil to 150 to 200 mg every 8 hours. Such dosages or higher ones (up to 400 mg every 8 hours) are generally reserved for severely ill patients, including those with thyroid storm, to block the conversion of T4 to T3. Maintenance dosing with propylthiouracil is 50 mg twice a day or 3 times a day.

Approximately 20 to 50% of patients with Graves disease remain in remission after a 1- to 2-year course of either methimazole or propylthiouracil (3). The return to normal or a marked decrease in gland size, the restoration of a normal serum TSH level, and resolution of hyperthyroid symptoms are good

prognostic signs of long-term remission. The concomitant use of antithyroid medications and [levothyroxine](#) does not improve the remission rate in patients with Graves disease. Because toxic nodular goiter rarely goes into remission, antithyroid medication is given only in preparation for surgical treatment or iodine-131 therapy.

Adverse effects include rash, allergic reactions, abnormal liver function (including hepatic failure with [propylthiouracil](#)), and, in approximately 0.1% of patients, reversible agranulocytosis. Patients allergic to one medication can be switched to the other, but cross-sensitivity may occur. If agranulocytosis occurs, the patient cannot be switched to the other medication; other therapy (eg, radioiodine, surgery) should be used.

Pearls & Pitfalls

- If agranulocytosis occurs with one of the antithyroid peroxidase medications (methimazole or propylthiouracil), avoid using another medication in the same class; use another therapy (eg, radioiodine, surgery) instead.

Potential adverse effects or other characteristics vary between the 2 medications and guide the indications for each. [Methimazole](#) need only be given once a day, which improves adherence. Furthermore, when methimazole is used in dosages of < 20 mg a day, agranulocytosis is less common; with [propylthiouracil](#), agranulocytosis may occur at any dosage.

[Methimazole](#) has been used successfully in patients who are pregnant or nursing without fetal or infant complications, but rarely methimazole has been associated with scalp and gastrointestinal defects in neonates and rarely other abnormalities. Because of these complications, [propylthiouracil](#) is used in the first trimester of pregnancy.

[Propylthiouracil](#) is preferred for the treatment of thyroid storm, because the high dosages used (over 800 mg a day) partially block the peripheral conversion of T4 to T3 in addition to decreasing production in the thyroid.

The combination of high-dose [propylthiouracil](#) and [dexamethasone](#), also a potent inhibitor of T4 to T3 conversion, can relieve symptoms of severe hyperthyroidism in patients with thyroid storm and restore the serum T3 level to normal within a week.

Beta-blockers

Symptoms and signs of hyperthyroidism due to adrenergic stimulation may respond to beta-blockers; [propranolol](#) has had the greatest use, but [atenolol](#) or [metoprolol](#) are alternatives.

Manifestations that typically respond to beta-blockers include:

- Tachycardia
- Tremor
- Some mental symptoms (eg, anxiety)

- Eyelid lag

Occasionally heat intolerance and sweating, diarrhea, and proximal myopathy also respond to beta-blockers.

Manifestations that typically do not respond to beta-blockers include:

- Goiter
- Exophthalmos
- Weight loss
- Bruit heard over the gland
- Increased oxygen consumption
- Increased circulating thyroxine levels

[Propranolol](#) is indicated in thyroid storm (see table [Treatment of Thyroid Storm](#)). It rapidly decreases heart rate, usually within 2 to 3 hours when given orally and within minutes when given intravenously. [Esmolol](#) should be used only in the intensive care unit because it requires careful titration and monitoring. Beta-blockers are also indicated for tachycardia with hyperthyroidism, especially in older patients, because antithyroid medications usually take several weeks to become fully effective.

Calcium channel blockers may control tachyarrhythmias in patients in whom beta-blockers are contraindicated.

Iodine

Iodine in pharmacologic doses inhibits the release of T3 and T4 within hours and inhibits the organification of iodine, a transitory effect lasting from a few days to a week, after which inhibition usually ceases. Iodine is used for emergency management of thyroid storm, for hyperthyroid patients undergoing emergency nonthyroid surgery, and (because it also decreases the vascularity of the thyroid) for preoperative preparation of patients with hyperthyroidism undergoing thyroidectomy. Iodine generally is not used for routine treatment of hyperthyroidism. The usual dosage is 2 to 3 drops (100 to 150 mg) of a saturated [potassium iodide](#) solution orally 3 times a day or 4 times a day or [sodium iodide](#) in 1 L 0.9% saline solution 0.5 to 1 g IV given slowly once a day.

Complications of iodine therapy include inflammation of the salivary glands, conjunctivitis, and rash.

Surgery

Surgery is indicated for patients with Graves disease whose hyperthyroidism has recurred after courses of antithyroid medications and who refuse iodine-131 therapy, patients who cannot tolerate antithyroid medications, patients with very large goiters, and in some younger patients with toxic adenoma and multinodular goiter. Surgery may be done in older patients with giant nodular goiters.

Surgery usually restores normal function. Postoperative recurrences vary between 2 and 16% ([4](#)); risk of hypothyroidism is directly related to the extent of surgery. Vocal fold paralysis and hypoparathyroidism are uncommon complications. Saturated solution of [potassium iodide](#) 3 drops (about 100 to 150 mg) orally 3 times a day should be given for 10 days before surgery to reduce the vascularity of the gland.

However, [methimazole](#) must be given first because the patient should be euthyroid before iodide is given. [Dexamethasone](#) and a beta-blocker can be added to [methimazole](#) to rapidly restore euthyroidism. Surgical procedures on the anterior neck are more difficult in patients who previously underwent thyroidectomy or radioiodine therapy because of scar tissue.

Treatment of thyroid storm

A treatment regimen for thyroid storm is shown in the table [Treatment of Thyroid Storm](#). The precipitating cause should also be addressed.

TABLE
Treatment of Thyroid Storm
Propylthiouracil : 600 mg orally given before iodine, then 200–400 mg every 6–8 hours
Iodine: 5 drops (0.25 mL) saturated solution of potassium iodide orally every 6 hours
or
4–8 drops Lugol's solution 2% orally every 6–8 hours
or
1 g sodium iodide slowly by intravenous drip over 24 hours
Propranolol : 60–80 mg orally every 4–6 hours with appropriate adjustment for heart rate and blood pressure
or
Propranolol : 0.5–1 mg intravenously over 10–15 minutes every 4 hours (not to exceed 1 mg/minute) under close monitoring
A repeat 1-mg propranolol dose given after 2 minutes, if needed, or esmolol
Intravenous dextrose solutions
Correction of dehydration and electrolyte imbalance
Cooling blanket for hyperthermia
Antiarrhythmics (eg, calcium channel blockers, adenosine, beta-blockers) if necessary for atrial fibrillation
Treatment of underlying disorder, such as infection
Glucocorticoids: Hydrocortisone 100 mg intravenously every 8 hours
or
Dexamethasone 2 mg intravenously every 6 hours

Definitive therapy after control of the crisis via ablation of the thyroid with iodine-131 or surgical treatment

Treatment of infiltrative dermopathy

In infiltrative dermopathy (in Graves disease), topical glucocorticoids or glucocorticoid injections into the lesions may decrease the dermopathy. Dermopathy sometimes remits spontaneously after months or years.

Treatment of infiltrative ophthalmopathy

Ophthalmopathy should be treated jointly by the endocrinologist and ophthalmologist and may require [selenium](#), glucocorticoids, orbital radiation, and surgery ([5](#)). Surgical thyroidectomy may help resolve or prevent progression of ophthalmopathy. [Teprotumumab](#), an insulin-like growth factor 1 (IGF-1) receptor inhibitor, is very effective therapy for moderately severe ophthalmopathy ([6](#)). Radioiodine therapy may accelerate progression of ophthalmopathy when ophthalmopathy is active, and is thus contraindicated in this active phase.

Management of subclinical hyperthyroidism

In patients with subclinical hyperthyroidism who are taking [levothyroxine](#), reduction of the dose is the most appropriate management unless therapy is aimed at maintaining suppressed TSH levels in patients with thyroid cancer.

Therapy is indicated for patients with endogenous subclinical hyperthyroidism (serum TSH < 0.1 mIU/L [< 0.1 microIU/mL]), especially those with atrial fibrillation or reduced bone mineral density. The usual treatment is iodine-131, but low doses of [methimazole](#) are also effective.

Treatment references

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Key Points

- The most common etiology of hyperthyroidism is Graves disease, caused by autoimmune-stimulated excessive hormone synthesis by the thyroid gland.
- Other causes of hyperthyroidism include excessive stimulation of a normal thyroid gland (eg, by thyroid-stimulating hormone [TSH], human chorionic gonadotropin [hCG], ingestion of iodine or iodine-containing medications), excessive hormone synthesis by an abnormal thyroid (eg, toxic nodular goiter), excessive release of thyroid hormones (eg, due to thyroiditis), or ingestion of excessive quantities of thyroid hormone.
- Symptoms and signs include tachycardia, fatigue, weight loss, nervousness, and tremor; patients with Graves disease may also have infiltrative ophthalmopathy and infiltrative dermopathy.
- Free thyroxine (T4), and/or free or total triiodothyronine (T3) are elevated, and TSH is suppressed (except in rarer cases of pituitary causes of hyperthyroidism).
- Hormone synthesis can be suppressed with [methimazole](#) (or in certain cases, [propylthiouracil](#)) and adrenergic symptoms relieved with beta-blockers; long-term treatment may require thyroid ablation with radioactive iodine or surgery.
- Thyroid storm—resulting from untreated or inadequately treated severe hyperthyroidism—is a life-threatening emergency that presents with severe symptoms of hyperthyroidism and may result in cardiovascular collapse or shock; it is treated with a regimen of antithyroid medications, iodine, beta-blockers, glucocorticoids, and hemodynamic support.

Drug Information for the Topic



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